TT1 Review

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vector space axioms

AI. $u+v \in V$

AII. $(u+v)+w = u+(v+w)$

AIII. $\exists\, 0 \in V$ such that $u+0 = u$

AIV. $\exists\ -u \in V$ such that $u+(-u) = 0$

MI. $au \in V$

MII. $a(bu) = (ab)u$

MIII. (a) $(a+b)u = au + bu$

(b) $a(u+v) = au + av$

MIV. $1u = u$

Cancellation theorem:

Let V be a vector space. If $u, v, w \in V$, then

if $u+v = u+w$,

then $v = w$

Prop I: for every u, $-u \in V$, $-u+u = 0$

Prop II: for every $u \in V$, $0+u = u$

Prop III: ① unique zero vector $0 \in V$

② for every $u \in V$, the negative $-u \in V$ is unique

③ $-(-u) = u$

Prop IV: for $u, v \in V$, $u+v = v+u$

Prop V: for all $u \in V$, $\alpha \in \mathbb{F}$:

① $0u = 0$

② $\alpha 0 = 0$

③ if $\alpha u = 0$, then either $\alpha = 0$ or $u = 0$

Prop VI: for all $u \in V$ and $\alpha \in \mathbb{F}$, $(-\alpha)u = -(\alpha u) = \alpha(-u)$

important facts

- every vector space is either infinite or $\{0\}$
- if $u \in V$ and $v \notin V$, then $u \neq v$ in V

Subspaces

subspace test, version 1

To show a subset U is a subspace of V:

subset & vector space

SI. $0 \in U$

SII. $u+v \in U$

SIII. $au \in U$

subspace test, version 2

To show a nonset U is a subspace of V, it suff $\alpha x+\beta y \in U$ for all $x, y \in U$ and $\alpha, \beta \in \mathbb{F}$,

then U is a subspace of V.

Linear combinations & span

linear combination

A vector $v \in V$ is a linear combination of $\{u_1, u_2, \dots u_n\} \subset V$ if and only if it

can be written as:

$$v = \sum_{i=1}^{n} a_i u_i \quad = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$$

for some $a_i \in \mathbb{F}$

span

$\operatorname{span}\{u_1, u_2, \dots u_n\} = \{v \mid v = \sum_{i=1}^{n} a_i u_i,\ \forall a_i \in \mathbb{F}\}$

linear combinations of $u_1, u_2, \dots u_n$

Prop I. span of $\{u_1, u_2, \dots u_n\} \subset V$ is a subspace of V

if S is a subset of vector space V, then $\operatorname{span} S$ is a subspace of V

Prop II. Let $U = \operatorname{span}\{u_1, u_2, \dots u_n\}$ which is a subspace of V.

If W is a subspace of V containing $\{u_1, u_2, \dots u_n\}$, then

U is a subspace of W.

span is a subspace of subspace of a vector space?

important subspaces

- image/column space of A: $\operatorname{im} A \triangleq \{y \mid y = Ax,\ x \in \mathbb{F}^n\} \subseteq \mathbb{F}^m$

$$\mathbb{F}^m \begin{bmatrix} 2 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbb{F}^n} = x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} 2 \\ 2 \end{bmatrix} + z\begin{bmatrix} 1 \\ 3 \end{bmatrix} \in \mathbb{F}^m$$

all vectors you can get when multiplying matrix by a vector

- null space of $A \in \mathbb{F}^{m \times n}$ (solution space): $\ker A \triangleq \{x \mid Ax = 0\} \subseteq \mathbb{F}^n$

$$\mathbb{F}^m \begin{bmatrix} 2 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbb{F}^n} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

all vectors that will produce 0 when multiplied by the matrix.

other facts

- If U and W are subspaces of vector space V, then $\operatorname{span}(U \cup W) = U + W$
- If U and W are subspaces, so is $U \cap W$

Linear dependence

$\{u_1, u_2, \dots u_n\} \subset V$ is linearly independent if and only if

$a_1u_1 + a_2u_2 + \dots + a_nu_n = 0$ implies that $a_1 = a_2 = \dots = a_n = 0$

Prop I. If $\{u_1, u_2, \dots u_n\} \subset V$ is LI, and $v = a_1u_1 + a_2u_2 + \dots + a_nu_n$,

then $a_1, a_2, \dots a_n$ is uniquely determined.

a linear combination of a set of linearly independent vectors is unique

Minimally spanning sets

Theorem. Let $\{u_1, u_2, \dots u_n\} \subset V$ where V is a vector space.

For every u_k with $k = 1, 2, \dots n$, (strictly a subset of, "$\subset$ but not $=$")

$\operatorname{span}\{u_1, \dots u_{k-1}, u_{k+1}, \dots u_n\} \subseteq \operatorname{span}\{u_1, \dots u_n\}$ iff

$\{u_1, \dots u_n\}$ is linearly independent

if a spanning set is LI, we cannot remove a vector without compromising the vector space → this is a minimally spanning set

Corollary. Let $\{u_1, u_2, \dots u_n\} \subset V$ where V is a vector space.

For at least one u_k, $\operatorname{span}\{u_1, u_2, \dots u_{k-1}, u_{k+1}, \dots u_n\} = \operatorname{span}\{u_1, \dots u_n\}$

iff $\{u_1, u_2, \dots u_n\}$ is linearly dependent.

if a spanning set is linearly dependent, we can remove a vector without affecting span

Fundamental theorem of linear algebra

Let V be a vector space spanned by n vectors. If a set of m vectors from V is linearly

independent, then $m \leq n$

Growing and pruning

Let $\{x_1, x_2, \dots x_n\}$ be a set of vectors in V

(a) suppose $\{x_1, \dots x_m\}$ LI and does not span V. If $x \in V$ and $x \notin \operatorname{span}\{x_1, \dots x_m\}$, then

$\{x_1, \dots x_m, x\}$ is LI

adding a linearly independent vector in a LI set will result in LI

(b) suppose $\{x_1, \dots x_n\}$ is linearly dependent and spans V. If $c_1x_1 + c_2x_2 + \dots + c_nx_n = 0$ is

a non-trivial linear combination, and $c_j \neq 0$ for some j, then $\{x_1, x_2, \dots x_{j-1}, x_{j+1}, \dots x_n\}$

spans V (a linearly dependent vector)

$x_j = -\frac{c_1}{c_j}x_1 - \frac{c_2}{c_j}x_2 - \dots - \frac{c_n}{c_j}x_n$

Replacement lemma

Suppose $\{x_1, x_2, \dots x_n\}$ spans V. Let x be a nonzero vector in V, with $x = c_1x_1 + c_2x_2 + \dots + c_nx_n$.

If $c_j \neq 0$, then the set with x_j replaced with x spans V.

$\operatorname{span}\{x_1, x_2, \dots x_{j-1}, x, x_{j+1}, \dots x_n\} = V$

you can replace a vector with a linear combination of the set that contains the vector